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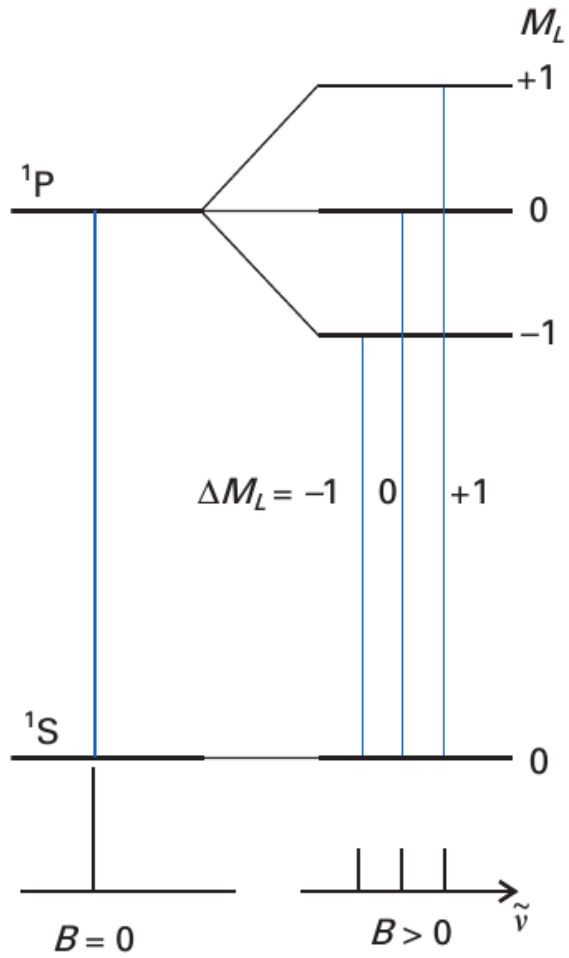
$$\boxed{E = E_n^{(0)} + \frac{eB}{2m} m_l \hbar = m_l \hbar \mu_B}$$

$\frac{e\hbar}{2mc} = \mu_B =$ Bohr magneton $= 9.247408 \times 10^{-28}\,\mathrm{J}\,T^{-1}$. (T $\equiv$ Tesla)

States for non-zero values of m and identical l no longer have same energy

This is why m is the magnetic quantum number.

Split in degeneracy due to magnetic field is called Normal Zeeman effect



Picture above is for an atom (not H) where the ground and excited states are singlets

When spin is considered, we get anomalous zeeman effect